SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
SIMATS ENGINEERING
COMPUTER SCIENCE AND ENGINEERING
CSA17 - ARTIFICIAL INTELLIGENCE

Experiment No. 12: Python Program to Implement Tic Tac Toe Game

Aim

To write a Python program to implement the Tic Tac Toe game that allows two players to play alternately on a 3x3 board, checking after every move whether a player has won or the game has ended in a draw.

Algorithm

1. Start.

2. Create a 3x3 board initialized with empty spaces, represented internally as a list of 9 elements.

3. Define a function display_board() to print the board in a 3x3 grid format.

4. Define a function check_winner(board, player) that checks all rows, columns and the two diagonals for three matching symbols belonging to the same player.

5. Define a function is_full(board) that checks whether every cell of the board has been filled (draw condition).

6. Set the current player to 'X'.

7. Repeat the following steps until the game ends: a) Display the board. b) Prompt the current player to enter a position (1-9). c) If the chosen position is empty, place the current player's symbol there; otherwise prompt again. d) Call check_winner() for the current player; if True, display the winner and end the game. e) Call is_full(); if True and there is no winner, declare a draw and end the game. f) Switch to the other player.

8. Stop.

Pseudocode

```
BEGIN
  CREATE board = [' '] * 9
  SET current_player = 'X'
  FUNCTION check_winner(board, player)
    FOR each winning combination (rows, columns, diagonals) DO
      IF all 3 positions in combination == player THEN
        RETURN True
      END IF
    END FOR
    RETURN False
  END FUNCTION
  WHILE True DO
    DISPLAY board
    INPUT position FROM current_player
    IF board[position] == ' ' THEN
      board[position] = current_player
    ELSE
```

```
      CONTINUE
    END IF
    IF check_winner(board, current_player) THEN
      PRINT current_player, 'wins'
      BREAK
    END IF
    IF board has no empty position THEN
      PRINT 'Game Draw'
      BREAK
    END IF
    SWITCH current_player between 'X' and 'O'
  END WHILE
END
```

Result

The Python program to implement the Tic Tac Toe game was executed successfully. Two players, X and O, played alternately by entering positions from 1 to 9. The board was updated and displayed after every move, and the program correctly detected the winning condition once player X completed a diagonal. Thus, the Tic Tac Toe game was implemented and verified successfully using Python.

Sample Output:

```
$ python3 tictactoe.py
 1 | 2 | 3
-----------
 4 | 5 | 6
-----------
 7 | 8 | 9

Player X, enter position (1-9): 1
Player O, enter position (1-9): 2
Player X, enter position (1-9): 5
Player O, enter position (1-9): 3
Player X, enter position (1-9): 9

 X | O | O
-----------
   | X |
-----------
   |   | X

Player X wins!
Game completed successfully!
```